

# A Bayesian perspective on some replacement strategies

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In this paper we present a Bayesian decision theoretic approach for determining optimal replacement strategies. This approach enables us to formally incorporate, express, and update our uncertainty when determining optimal replacement strategies. We develop relevant expressions for both the block replacement protocol with minimal repair and the age replacement protocol and illustrate the use of our approach with real data. © 1996 Elsevier Science Limited.

## INTRODUCTION AND OVERVIEW

It is common practice to use planned replacement strategies for systems (or components) that experience aging. The underlying assumption behind such strategies is that an in-service failure of the system (and the subsequent replacement) is more costly than a planned replacement. The optimal strategies consider the trade-off between these two costs. Most of the approaches presented in the literature are based on the assumption that the failure characteristics (failure model) of the system are known and thus the replacement strategies are dependent on such failure characteristics.<sup>1,2</sup> Exceptions to these are the works of Sathe & Hancock,<sup>3</sup> Bassin,<sup>4</sup> and Fox.<sup>5</sup> Both Refs 3 and 5 consider the case of age replacement policies when a Weibull distribution is the underlying failure model. Sathe & Hancock<sup>3</sup> allude to a Bayesian analysis by considering prior distributions on the shape and scale parameters of the Weibull model and attempt to develop a Bayesian analog of the classical solution to the problem by minimizing the expected long-run average cost. Fox<sup>5</sup> considers the same structure as in 3 with the additional assumption that shape parameter is known. The author in Ref. 5 uses a discounted cost structure as a loss function and analyzes the problem of adaptive replacement

strategies via the updating of the prior distribution for the scale parameter. Asymptotic results are obtained for an infinite planning horizon. Bassin<sup>4</sup> considers the block replacement policy when only minimal repair is made at failure by using Bayesian point estimates for the parameters of the Weibull process.

In this article we present a fully Bayesian analysis of the optimal replacement problem for both the block replacement protocol with minimal repair and the age replacement protocol. We develop a Bayesian decision theoretic formulation of the problem by defining a cost-based utility function and describing our uncertainty about underlying failure characteristics of the system (or component) probabilistically. This formulation provides a coherent framework where optimal replacement strategies are obtained by maximizing the expected utility. It also allows us to quantify risk associated with different replacement strategies via probability statements concerning cost per unit time for a given strategy. Finally, the approach enables us to revise our uncertainty as we learn about failure behavior of the system, and to adapt our replacement strategies accordingly.

In the following section, we give an overview of the block and age replacement protocols. In considering the block replacement protocol with minimal repair, we assume that the failure behavior of the system can

be described by a nonhomogeneous Poisson process with a Weibull intensity function. We assume a Weibull failure model for illustrating the age replacement protocol. In the next section, we present the Bayesian formulation of the optimal replacement problem and develop strategies for both replacement protocols based on the minimization of expected cost per unit time. Furthermore, we develop probability distributions for cost per unit time and discuss the revision of uncertainties as failure/replacement data becomes available under each protocol. We illustrate the approaches with actual data. In the last section we present some closing comments on our approach.

## REPLACEMENT MODELS

There are two well known protocols for replacement, block replacement and age replacement.<sup>6</sup> For completeness and introduction of notation, we give a brief overview of the models for these protocols.

### Block replacement protocol

Under the usual block replacement protocol, we assume that a planned replacement will be made at time epochs  $t_B, 2t_B, \dots$ , etc., irrespective of the age of the system and that an in-service replacement is made whenever the system fails. Herein we consider an adaptation of this protocol introduced by Barlow & Hunter<sup>7</sup> whereby items are minimally repaired upon failure but replaced at times  $t_B, 2t_B, \dots$ , etc. Minimal repair means that the item can be repaired so that its failure characteristics (reliability, failure rate, etc.) are as they were just prior to failure. In what follows we refer to this as the block replacement protocol with minimal repair. The original block replacement protocol can also be considered in the Bayesian framework but requires the estimation of the renewal function and is thus more computationally intensive. The complexity of this problem can be reduced by using results on approximation of the renewal function such as those suggested by Smith & Leadbetter<sup>8</sup> and more recently by Smeitink & Dekker.<sup>9</sup>

To introduce some notation, let  $c_P$  denote the cost of a planned replacement and  $c_F$  denote cost of an in-service repair such that  $c_F > c_P$ . Furthermore, let  $N(t)$  denote the number of failures in an interval of length  $t$ . In a planned replacement cycle of length  $t_B$ , the cost is thus given by

$$c_P + c_F N(t_B)$$

and the cost per unit time is given as

$$C(t_B) = \frac{c_P + c_F N(t_B)}{t_B}. \quad (1)$$

The decision problem, then, is to find  $t_B$  such that  $C(t_B)$  is minimized.

We note that  $N(t_B)$  is the only random quantity in (1). Making the usual assumption of aging, we model the failure process by assuming that  $N(t)$  is a nonhomogeneous Poisson process with intensity function

$$m(t) = \alpha \beta t^{\beta-1} \quad (2)$$

where  $\alpha$  and  $\beta$  are unknown coefficients. This is a common assumption in the reliability literature (see for example Lee & Lee<sup>10</sup> and Kyparisis & Singpurwalla)<sup>11</sup>. Since  $m(t)$  resembles the failure rate of the well-known Weibull distribution, this process is often referred to as the Weibull process.

Given the above intensity function and the assumption of minimal repair upon failure, the probability of  $n$  in-service failures in an interval of length  $t_B$  is given as

$$\Pr\{N(t_B) = n\} = \frac{\{M(t_B)\}^n}{n!} \exp\{-M(t_B)\} \quad (3)$$

where

$$M(t) = E[N(t) | \alpha, \beta] = \int_0^t m(t) dt = \alpha t^\beta \quad (4)$$

is referred to as the Mean Value Function. Note also from (3) and (4) that the distribution for  $T_1$ , the time to first failure, can be written as

$$\Pr\{T_1 \geq t | \alpha, \beta\} = \Pr\{N(t) = 0 | \alpha, \beta\} = \exp\{-\alpha t^\beta\} \quad (5)$$

which is the reliability function for the Weibull failure model used extensively in reliability and maintenance analysis.

The common approach in the literature for developing an optimal block replacement strategy is to assume that  $\alpha$  and  $\beta$  are given (or estimated based on past data) and to select the block replacement interval,  $t_B^*$ , which minimizes the expected cost per unit time.

$$E[C(t_B) | \alpha, \beta] = E_N \left[ \frac{c_P + c_F N(t_B)}{t_B} \middle| \alpha, \beta \right] = \left[ \frac{c_P + c_F \alpha t_B^\beta}{t_B} \right]. \quad (6)$$

Assuming that  $\beta > 1$  (which implies aging) we obtain

$$t_B^* = \left( \frac{c_P}{c_F \alpha (\beta - 1)} \right)^{1/\beta} \quad (7)$$

which is a function of both  $\alpha$  and  $\beta$ .

### Age replacement protocol

Under the age replacement protocol, we assume that a planned replacement is made when the age of the

system reaches a specified time,  $t_A$ , or an in-service replacement is made whenever the system fails. Defining the replacement cycle as the time between successive replacements (either planned or in-service) and using the notation defined above, we develop the cost per unit time along the following lines.

Note that length of the replacement cycle is a random variable defined as

$$L(T, t_A) = \begin{cases} t_A & \text{if } T \geq t_A \\ T & \text{if } T < t_A \end{cases} \quad (8)$$

and the cost per replacement cycle is also a random variable given by

$$Q(T, t_A) = \begin{cases} c_P & \text{if } T \geq t_A \\ c_F & \text{if } T < t_A \end{cases} \quad (9)$$

where  $T$  is the time to failure which we model here using the Weibull density

$$f(t | \alpha, \beta) = \alpha \beta t^{\beta-1} \exp\{-\alpha t^\beta\}. \quad (10)$$

Thus, the cost per unit time is given as

$$C(T, t_A) = \begin{cases} c_P/t_A & \text{if } T \geq t_A \\ c_F/T & \text{if } T < t_A \end{cases}. \quad (11)$$

The common approach in literature for developing an optimal age replacement strategy is to assume that  $\alpha$  and  $\beta$  are given (or estimated based on past data) and to select the replacement age,  $t_A^*$ , which minimizes the expected cost per unit time. However, rather than considering the minimization  $E[C(T, t_A)]$  directly, it is common to invoke the Renewal Reward Theorem (see, for example, Cox<sup>12</sup>) and then minimize

$$\begin{aligned} & \frac{E_T[Q(T, t_A) | \alpha, \beta]}{E_T[L(T, t_A) | \alpha, \beta]} \\ &= \frac{c_P R(t_A | \alpha, \beta) + c_F F(t_A | \alpha, \beta)}{t_A R(t_A | \alpha, \beta) + \int_0^{t_A} t f(t | \alpha, \beta) dt} \end{aligned} \quad (12)$$

where  $R(t | \alpha, \beta) = \int_t^\infty f(x | \alpha, \beta) dx = \exp\{-\alpha t^\beta\}$  and  $F(t | \alpha, \beta) = 1 - R(t | \alpha, \beta)$ . This minimization involves numerical procedures for evaluating the integral in the denominator. It is clear from the above development, however, that  $t_A^*$  will be a function of  $\alpha$  and  $\beta$ .

## DEVELOPMENT OF BAYESIAN REPLACEMENT STRATEGIES

To develop optimal replacement strategies under the Bayesian paradigm we consider a decision theoretic set up of the problem. This entails representation of all uncertain quantities via probability distributions

and the specification of an appropriate utility function for making the decision. The Bayesian solution, then, is obtained via maximization of the expected utility with respect to  $t_B$  (block replacement protocol with minimal repair) or  $t_A$  (age replacement protocol), respectively. If economics is the only concern in the development of the optimal strategy, then an appropriate utility function is the negative of the cost per unit time. We proceed under this assumption and thus the maximization of the expected utility is obtained as the minimization of expected cost per unit time. The expectation is taken over all random quantities including the unknown parameters.

For both the age replacement protocol and the block replacement protocol with minimal repair, the underlying model for the time to first failure is the Weibull density given in (10). Thus, for both protocols, our analysis involves expressing our uncertainty about the unknown coefficients  $\alpha$  and  $\beta$  via prior distributions. We select, as an appropriate prior for  $\alpha$ , the gamma distribution given by

$$f(\alpha) = \frac{b^a}{\Gamma(a)} \alpha^{a-1} e^{-b\alpha}, \quad \alpha > 0 \quad (13)$$

denoted  $\alpha \sim G(a, b)$ , where  $a, b > 0$  are specified parameters. For the prior distribution for the shape parameter  $\beta$ , it is customary to define a discrete distribution as in Soland.<sup>13</sup> We define a discrete distribution by using a discretization of the beta density on  $(\beta_L, \beta_U)$  since this allows for great flexibility in representing prior uncertainty. The beta density is given by

$$g(\beta) = \frac{\Gamma(c+d)}{\Gamma(c)\Gamma(d)} \frac{(\beta - \beta_L)^{c-1} (\beta_U - \beta)^{d-1}}{(\beta_U - \beta_L)^{c+d-1}} \quad \text{for } 0 \leq \beta_L \leq \beta \leq \beta_U \quad (14)$$

where  $\beta_L, \beta_U, c, d > 0$  are specified parameters. We define our distribution for  $\beta$  as

$$P_l = \Pr\{\beta = \beta_l\} = \int_{\beta_l - \delta/2}^{\beta_l + \delta/2} g(\beta) d\beta \quad (15)$$

where  $\beta_l = \beta_L + \delta(2l-1)/2$  and  $\delta = (\beta_U - \beta_L)/k$  for  $l = 1, \dots, k$ . We further assume that, initially, the quantities  $\alpha$  and  $\beta$  are independent and thus the joint prior distribution is given as the product of (13) and (15). We note that alternative forms for the prior joint distribution of  $\alpha$  and  $\beta$  are also possible.

The selection of prior parameter values is based on uncertainty about both the aging characteristics and the time until first failure. For example,  $\beta$  controls the rate at which the component ages and for items which experience aging as discussed here,  $\beta > 1$ . A reasonable upper bound for  $\beta$  can also be constructed. The parameters  $c$  and  $d$  of (14) can be determined as usual by eliciting a best guess value and an uncertainty

interval for  $\beta$ . Once these are available, the values of  $c$  and  $d$  may be obtained by equating the elicited values to appropriate moment/probability expressions from (14) and solving these equalities for  $c$  and  $d$ . Alternatively, choosing  $c = d = 1$  in (14) yields a uniform prior for  $\beta$  on the interval  $(\beta_L, \beta_U)$ .

Once the prior distribution for  $\beta$  has been established, the parameters of (13) may be obtained in a similar manner by eliciting information about the time to first failure and equating these with appropriate expressions from the predictive distribution

$$f(t) = E_{\alpha, \beta}[f(t | \alpha, \beta)] = \sum_{i=1}^k \frac{ab^a \beta_i t^{\beta_i-1}}{(b + t^{\beta_i})^{a+1}} P_i.$$

The framework provided by the Bayesian formulation allows a much more in depth study of the selection of an 'optimal' replacement strategy. Specifically, the uncertainty about the parameters captured in the prior/posterior distributions enables us to evaluate the risk associated with any replacement strategy via probability statements concerning the cost of the replacement strategy. We note that the prior distribution of  $\alpha$  and  $\beta$  can be updated with failure and replacement data as it becomes available thus allowing for revision of our replacement strategies in light of new evidence.

#### Bayesian strategy for the block replacement protocol with minimal repair

The optimal strategy for the block replacement model with minimal repair is obtained as the value  $t_B$  which minimizes the expected cost

$$\begin{aligned} E[C(t_B)] &= E_{\alpha, \beta}[E_T[C(t_B) | \alpha, \beta]] \\ &= \sum_{i=1}^k \frac{c_P + c_F \left(\frac{a}{b}\right) t_B^{\beta_i}}{t_B} P_i. \end{aligned} \quad (16)$$

The minimum is easily obtained with numerical techniques.

Note that a comparison of (6) and (16) reveals the difference in the classical and Bayesian approaches. A classical approach would require replacing  $\alpha$  and  $\beta$  by specific values, say  $(\hat{\alpha}, \hat{\beta})$ , while the Bayesian approach explicitly accounts for the uncertainty in  $(\alpha, \beta)$  via a joint probability distribution. That is, (16) represents a mixture of curves obtained from (6) for selected values of  $\alpha$  and  $\beta$ . As this mixture is a function of the uncertainty in  $(\alpha, \beta)$ , the two approaches tend to agree as the joint distribution converges to a degenerate distribution at  $(\hat{\alpha}, \hat{\beta})$ .

Probability statements concerning  $C(t_B)$  can be made for a fixed interval size,  $t_B$ , from (1) as

$$\begin{aligned} \Pr\{C(t_B) \leq C^*\} &= E_{\alpha, \beta}[\Pr\{C(t_B) \leq C^* | \alpha, \beta\}] \\ &= E_{\alpha, \beta}\left[\Pr\left\{N(t_B) \leq \frac{C^* t_B - c_P}{c_F} \mid \alpha, \beta\right\}\right] \\ &= \sum_{i=1}^k \left\{ \sum_{n \leq (c^* t_B - c_P)/c_F} \frac{\Gamma(a+n)}{\Gamma(a)n!} \right. \\ &\quad \left. \times \left(\frac{b}{b + t_B^{\beta_i}}\right)^a \left(1 - \frac{b}{b + t_B^{\beta_i}}\right)^n \right\} P_i. \end{aligned} \quad (17)$$

To develop the concept of an adaptive replacement strategy, we consider a single system which is minimally repaired upon failure but replaced by a new system at the end of the replacement cycle  $t_B$ . At the end of each replacement cycle, we revise our uncertainty about  $\alpha$  and  $\beta$  in the light of data observed. For example, if during the block interval of size  $t_B$ ,  $n$  system failures were observed at times  $0 < t_1 < \dots < t_n < t_B$ , then we may write the likelihood as

$$\mathcal{L}(\alpha, \beta) = \left\{ \prod_{i=1}^n \alpha \beta_i t_i^{\beta_i-1} \right\} \exp\{-\alpha t_B^{\beta_i}\}$$

where  $\prod_{i=1}^0 \{ \cdot \} \equiv 1$  will account for the case of no failures. The posterior distribution of  $\alpha$  and  $\beta$ , given  $t^{(n)} = (t_1, \dots, t_n)$ , is obtained via Bayes theorem as

$$\begin{aligned} f(\alpha, \beta_i | t^{(n)}) &\propto \left\{ \prod_{i=1}^n \alpha \beta_i t_i^{\beta_i-1} \right\} \\ &\quad \times \exp\{-\alpha t_B^{\beta_i}\} \alpha^{a-1} \exp\{-b\alpha\} P_i \end{aligned}$$

and after rearranging terms we obtain the posterior joint distribution of  $\alpha$  and  $\beta$  as

$$\frac{\alpha^{n+a-1} \beta_i^n \left\{ \prod_{j=1}^n t_j \right\}^{\beta_i-1} \exp\{-\alpha(b + t_B^{\beta_i})\} P_i}{\sum_{i=1}^k P_i \beta_i^n \left\{ \prod_{j=1}^n t_j \right\}^{\beta_i-1} \Gamma(n+a)/(b + t_B^{\beta_i})^{n+a}} \quad (18)$$

Note that the above can be written as

$$f(\alpha, \beta_i | t^{(n)}) = f(\alpha, | \beta_i, t^{(n)}) \Pr\{\beta = \beta_i | t^{(n)}\}$$

where  $(\alpha, | \beta_i, t^{(n)}) \sim G(a^*, b^*)$  with  $a^* = a + n$ ,  $b^* = b + t_B^{\beta_i}$ , and the posterior distribution of  $\beta$  can be obtained as

$$\begin{aligned} \Pr\{\beta = \beta_i | t^{(n)}\} &= P_i^* \\ &= \frac{\beta_i^n \left\{ \prod_{j=1}^n t_j \right\}^{\beta_i-1} / (b + t_B^{\beta_i})^{n+a}}{\sum_{i=1}^k P_i \beta_i^n \left\{ \prod_{j=1}^n t_j \right\}^{\beta_i-1} / (b + t_B^{\beta_i})^{n+a}} P_i. \end{aligned} \quad (19)$$

Thus the posterior distribution of  $\alpha$  and  $\beta$  are no longer independent. Revision of the optimal replacement strategies as well as the probability distribution

for the cost per unit time is achieved in a straight forward manner by replacing the prior quantities  $a$ ,  $b$  and  $P_i$  by the posterior quantities  $a^*$ ,  $b^*$  and  $P_i^*$ , respectively, in eqns (16) and (17).

#### An illustration of the block replacement protocol with minimal repair

We apply this methodology to the airplane air conditioner data (the failure times of air conditioners from 10 airplanes which undergo repair) of Proschan.<sup>14</sup> This data was shown to be reasonably modeled by the Weibull process by Bain & Engelhardt<sup>15</sup> among others. Initially, and for illustrative purposes only, we select our prior distributions for  $\alpha$  and  $\beta$  using (13)–(15) with  $a = 2.1$ ,  $b = 3$ ,  $c = 2$ ,  $d = 2$ ,  $\beta_L = 1$ , and  $\beta_U = 3$ . Additionally, assuming  $c_P = \$500$  and  $c_F = \$1000$ , yields an optimal replacement interval of  $t_B^* = 0.8574$  in 1000s of operating hours (or approximately 857.4 hours of operation) at an expected cost of \$1184.76 per 1000 hours of operation.

We employ the above replacement strategy for the air conditioner from plane #7914 in Ref. 15 and record the times of successive failures for this unit until 857.4 operating hours have been logged. These are given in Table 1 as cycle 1 failure times. The first failure occurred at 50 hours of operation and the second (after minimal repair) occurred at 94 hours of operation, implying that the unit survived for an additional 44 hours due to repair. During this cycle, the air conditioning unit failed and was minimally repaired 11 times. At 857.4 hours of operation, the strategy requires a replacement of the air conditioner (before the 12th failure occurs) signifying the end of cycle 1.

At this point, we update our parameters based on the failure data from cycle 1 and again determine the optimal replacement interval for a new air conditioner. The data for the new air conditioner is obtained from another airplane and the process is repeated. The data for cycle 2 is obtained from plane #7915 and that for cycle 3 is obtained from plane #7909. It should be noted that these three airplanes were selected due to the similarity of the maximum

likelihood estimates for  $\beta$  obtained in Ref. 15. Thus the air conditioners considered were as close to being statistically identical as possible.

We illustrate the adaptive nature of our approach by considering three cycles and at the end of each cycle, the failure data is used to determine the new optimal replacement interval. This information is summarized in Table 1 where column 1 shows the cycle number and column 2 shows the successive failure times in a cycle. Columns 3, 4, 5, respectively, show the expected number of failures in the first 1000 operating hours, the optimal replacement interval, and the minimum expected cost per unit time posterior to the data obtained up to and including the given cycle. Note that from (4),  $E[N(1)] = E[\alpha]$  is an indication of the rate at which the air conditioners deteriorate after repair for a unit time of operation (1000 hours).

In Fig. 1, we compare the prior distribution and the posterior distributions of  $\alpha$  and  $\beta$  after each replacement cycle. Note that there is a shift in the distribution of  $\beta$  to the left and a corresponding shift of the distribution of  $\alpha$  to the right as data becomes available. Furthermore, as data accumulates, the distribution for  $\beta$  appears to stabilize and become more peaked while the distribution of  $\alpha$  has become more diffuse. This latter result is due to the rather peaked shape of the prior distribution for  $\alpha$  indicating a strong prior certainty in the value of  $\alpha$  that was not reinforced by the data.

The revision of our uncertainty concerning  $\alpha$  and  $\beta$  after each cycle results in a revision of our expected cost curve. This requires a reassessment of our optimal replacement interval from the previous cycle in order for our replacement strategy to remain optimal with respect to the revised expected cost curve. The adaptive nature of our approach is illustrated in Fig. 2 which represents plots of (16), the expected cost curve as a function of the replacement interval  $t_B$ , and (17), the cumulative cost distribution at the optimal replacement interval  $t_B^*$ , a priori and after each cycle. Note the upward shift in the expected cost curves as well as a left shift in the optimal replacement interval. In addition, the cumulative cost distribution shifts to the right. These results are due to our prior optimism about the deterioration rate of the

Table 1. Summary of block replacement protocol for airplane air conditioners

| Cycle | Failure times†                                                                         | $E[N(1)]$ | $t_B^*$ | $E[C(t_B^*)]‡$ |
|-------|----------------------------------------------------------------------------------------|-----------|---------|----------------|
| 0     | —                                                                                      | 0.70      | 0.857 4 | 1 184.76       |
| 1     | 0.050, 0.094, 0.196, 0.268, 0.290, 0.329,<br>0.332, 0.347, 0.544, 0.732, 0.811, 0.899§ | 3.43      | 0.579 3 | 3 765.69       |
| 2     | 0.359, 0.368, 0.380, 0.650§                                                            | 3.75      | 0.521 3 | 3 983.02       |
| 3     | 0.090, 0.100, 0.160, 0.346, 0.407, 0.456<br>0.470, 0.494, 0.550§                       | 5.08      | 0.451 6 | 5 160.37       |

§ Replaced, † in 1000s of operating hours, ‡ in \$/1000 operating hours.

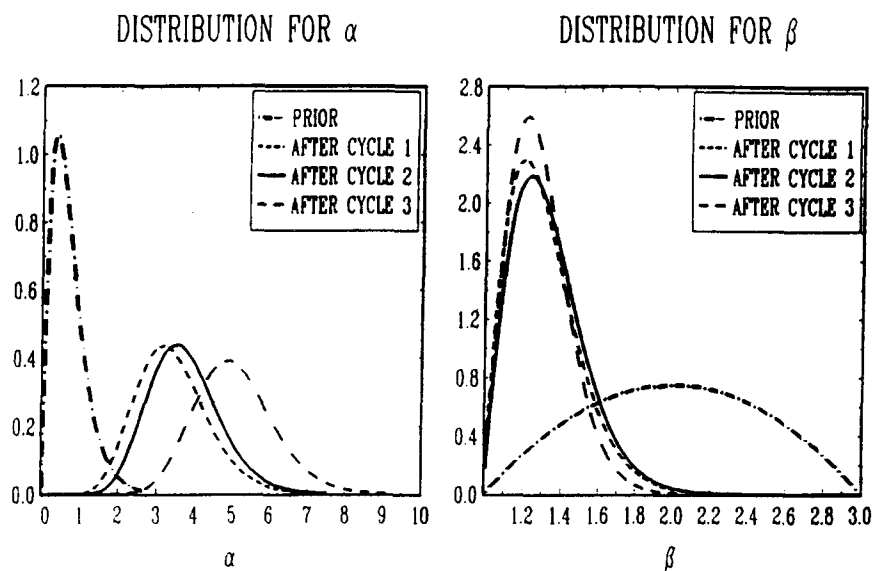


Fig. 1. Prior and posterior distributions of  $\alpha$  and  $\beta$  for block replacement with minimal repair.

air conditioning units which is not reflected in the failure data. This revision of the degree of our optimism is further illustrated by the consistent increase of  $E[N(1)]$  in Table 1.

The cumulative cost distribution can be used to provide a risk profile for the optimal replacement strategy. For example, note from Fig. 2 that there is a significant probability that the cost will exceed the

optimal expected cost.

#### Bayesian strategy for the age replacement protocol

The optimal strategy for the age replacement model is obtained as the value  $t_A$  which minimizes  $E_{\alpha,\beta}[E[C(T, t_A) | \alpha, \beta]]$ . Here, however, unlike (12), we do not rely on the Renewal Reward Theorem but rather calculate the inside expectation directly. From

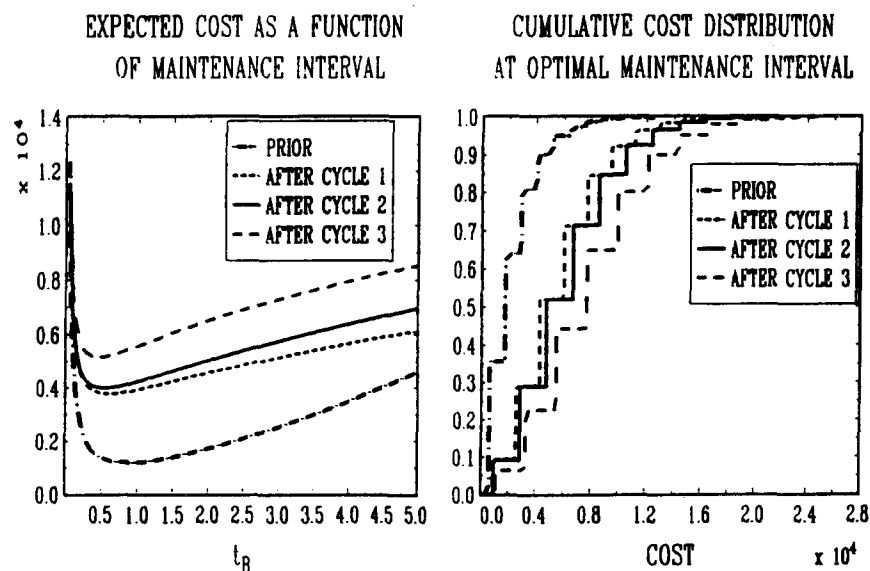


Fig. 2. Prior and posterior expected cost curves and cumulative cost distributions for block replacement with minimal repair.

(11) we write

$$\begin{aligned} E_T[C(T, t_A | \alpha, \beta)] &= \int_0^{t_A} (c_F/t)f(t | \alpha, \beta) dt \\ &\quad + \int_{t_A}^{\infty} (c_P/t_A)f(t | \alpha, \beta) dt \\ &= c_F \int_0^{t_A} \alpha \beta t^{\beta-2} \exp\{-\alpha t^\beta\} dt \\ &\quad + (c_P/t_A) \exp\{-\alpha t_A^\beta\}. \end{aligned} \quad (20)$$

Thus  $E_{\alpha, \beta}[E_T[C(T, t_A | \alpha, \beta)]]$  is given as

$$\sum_{i=1}^k \left[ \int_0^{\infty} \left\{ c_F \int_0^{t_A} \alpha \beta_i t^{\beta_i-2} \exp\{-\alpha t^{\beta_i}\} dt + (c_P/t_A) \exp\{-\alpha t_A^{\beta_i}\} \right\} \frac{b^a}{\Gamma(a)} \alpha^{\alpha-1} e^{-b\alpha} d\alpha \right] P_i.$$

Switching the order of integration and using integration by parts we obtain

$$\begin{aligned} E_{\alpha, \beta}[E_T[C(T, t_A | \alpha, \beta)]] &= \sum_{i=1}^k \left\{ c_F \int_0^{t_A} \frac{ab^a \beta_i t^{\beta_i-2}}{(b + t^{\beta_i})^{a+1}} dt \right. \\ &\quad \left. + (c_P/t_A) \left( \frac{b}{b + t_A^{\beta_i}} \right)^a \right\} P_i \end{aligned} \quad (21)$$

where the single dimensional integral is easily evaluated using numerical techniques. Thus the optimal replacement interval is determined as that which minimizes (21).

Probability statements concerning  $C(T, t_A)$  can be made from (11) as

$$\begin{aligned} \Pr\{C(T, t_A) \leq C^*\} &= E_{\alpha, \beta}[\Pr\{C(T, t_A) \leq C^* | \alpha, \beta\}] \\ &= E_{\alpha, \beta}[\Pr\{C(T, t_A) \leq C^* | \alpha, \beta, T \leq t_A\} \\ &\quad \times \Pr\{T \leq t_A | \alpha, \beta\} \\ &\quad + E_{\alpha, \beta}[\Pr\{C(T, t_A) \leq C^* | \alpha, \beta, T > t_A\} \\ &\quad \times \Pr\{T > t_A | \alpha, \beta\}] \\ &= E_{\alpha, \beta}[\Pr\{c_F/C^* \leq T \leq t_A | \alpha, \beta\} \\ &\quad + \Pr\{c_P/t_A \leq C^* | \alpha, \beta, T > t_A\} \\ &\quad \times R(t_A | \alpha, \beta)]. \end{aligned}$$

The above may be written more formally as

$$\begin{aligned} E_{\alpha, \beta}[I\{c_F/t_A \leq C^*\}(R(c_F/C^* | \alpha, \beta) \\ - R(t_A | \alpha, \beta)) + I\{c_P/t_A \leq C^*\}R(t_A | \alpha, \beta)] \end{aligned}$$

where  $I\{\cdot\}$  is the indicator function. Taking the expectation with respect to  $\alpha$  and  $\beta$  we obtain

$$\begin{aligned} \Pr\{C(T, t_A) \leq C^*\} &= \sum_{i=1}^k \{I(c_F/t_A \leq C^*)\} \\ &\quad \times \left[ \left( \frac{b}{b + (c_F/C^*)^{\beta_i}} \right)^a - \left( \frac{b}{b + t_A^{\beta_i}} \right)^a \right] \\ &\quad + I\{c_P/t_A \leq C^*\} \left( \frac{b}{b + t_A^{\beta_i}} \right)^a P_i. \end{aligned} \quad (22)$$

To develop an adaptive replacement strategy for the age replacement protocol, we assume that we will revise our uncertainty about  $\alpha$  and  $\beta$  at the end of an age replacement cycle which may either be at a failure or  $t_A$ . The likelihood function of  $\alpha$  and  $\beta$  can be written as

$$\begin{aligned} \mathcal{L}(\alpha, \beta) &= \alpha \beta t^{\beta-1} \exp\{-\alpha t^\beta\} I\{t \leq t_A\} \\ &\quad + \exp\{-\alpha (t_A)^\beta\} I\{t > t_A\}. \end{aligned}$$

The posterior distribution of  $\alpha$  and  $\beta$  is again obtained via Bayes Theorem as

$$f(\alpha, \beta_i | I\{t \leq t_A\}) = \begin{cases} \frac{\alpha^a \beta_i t^{\beta_i-1} \exp\{-\alpha (b + t^{\beta_i})\} P_i}{\sum_{i=1}^k P_i \beta_i t^{\beta_i-1} \Gamma(a+1) / (b + t^{\beta_i})^{a+1}} & \text{if } t \leq t_A \\ \frac{\alpha^{a-1} \exp\{-\alpha (b + t_A^{\beta_i})\} P_i}{\sum_{i=1}^k P_i \Gamma(a) / (b + t_A^{\beta_i})^a} & \text{if } t > t_A \end{cases} \quad (23)$$

where the posterior conditional distribution of  $(\alpha | \beta_i)$  is  $G(a^*, b^*)$  with  $a^* = a + 1$  and  $b^* = b + t^{\beta_i}$  when a failure was observed and  $a^* = a$  and  $b^* = b + t_A^{\beta_i}$  for a planned replacement. From (23) the posterior distribution of  $\beta$  is

$$P_i^* = \begin{cases} \frac{\beta_i t^{\beta_i-1} / (b + t^{\beta_i})^{a+1}}{\sum_{i=1}^k P_i \beta_i t^{\beta_i-1} / (b + t^{\beta_i})^{a+1}} P_i & \text{if } t \leq t_A \\ \frac{1 / (b + t_A^{\beta_i})^a}{\sum_{i=1}^k P_i / (b + t_A^{\beta_i})^a} P_i & \text{if } t > t_A \end{cases} \quad (24)$$

Revision of the optimal replacement strategies as well as the probability distribution for the cost per unit time follows along the lines presented for the block replacement protocol with minimal repair.

### An illustration of the age replacement protocol

We apply the Bayesian age replacement approach to the airplane air conditioner data in Ref. 15. We select our prior parameters for  $\alpha$  and  $\beta$  as in the block replacement with minimal repair illustration. We assume, however, that  $c_P = \$200$  and  $c_F = \$1000$ .

**Table 2. Summary of age replacement protocol for airplane air conditioners**

| Cycle | Failure times† | $t_A^*$ | $E[C(T, t_A^*)]‡$ |
|-------|----------------|---------|-------------------|
| 0     | —              | 0.442 5 | 2 040.33          |
| 1     | 0.050          | 0.321 7 | 5 900.82          |
| 2     | 0.3598         | 0.335 6 | 5 310.36          |
| 3     | 0.090          | 0.265 4 | 9 238.73          |

§ Replaced † in 1000s of operating hours. ‡ in \$/1000 operating hours.

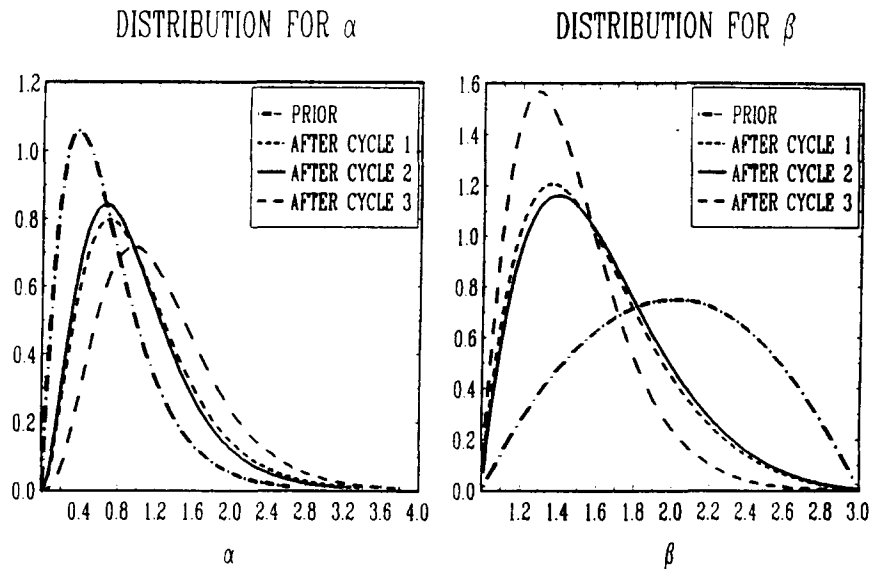


Fig. 3. Prior and posterior distributions of  $\alpha$  and  $\beta$  for age replacement.

Based on the prior assessments, the estimated optimal replacement strategy for the age replacement protocol is  $t_A^* = 0.4425$  in 1000s of operating hours (or 442.5 hours of operation) at an expected cost of \$2,040.33 per 1000 hours of operation.

Because in this model we assume replacement rather than minimal repair at failure, we use only the first failure data from each air conditioner. For each airplane air conditioner, the first failure is considered an in-service failure only if the failure time is less than  $t_A^*$ . If the failure time is greater than  $t_A^*$ , then we consider the air conditioner to be replaced at  $t_A^*$  before failure. As in the previous illustration, we use only three replacement cycles and thus again use the data from airplanes #7914, #7915, and #7909. For example, in cycle 1, our prior assessments require that

$t_A^* = 0.4425$  and so the air conditioner for airplane #7914 fails at  $t = 0.05 < t_A^*$  and is replaced by the new air conditioner from airplane #7915. Upon the revision of our uncertainty (after the first cycle) our assessments require that  $t_A^* = 0.3217$  and at this time, the air conditioner for airplane #7915 is still in operation (since the recorded failure time is  $t = 0.359 > t_A^*$ ) and a planned replacement occurs. The data is summarized in Table 2.

In Fig. 3, we compare the prior distribution and the posterior distributions of  $\alpha$  and  $\beta$  after each age replacement cycle. Note the results here are similar to those discussed in the previous illustration, however, both the distributions for  $\alpha$  and  $\beta$  are more diffuse due to the smaller amount of data obtained in each replacement cycle. In Fig. 4 we present the prior and

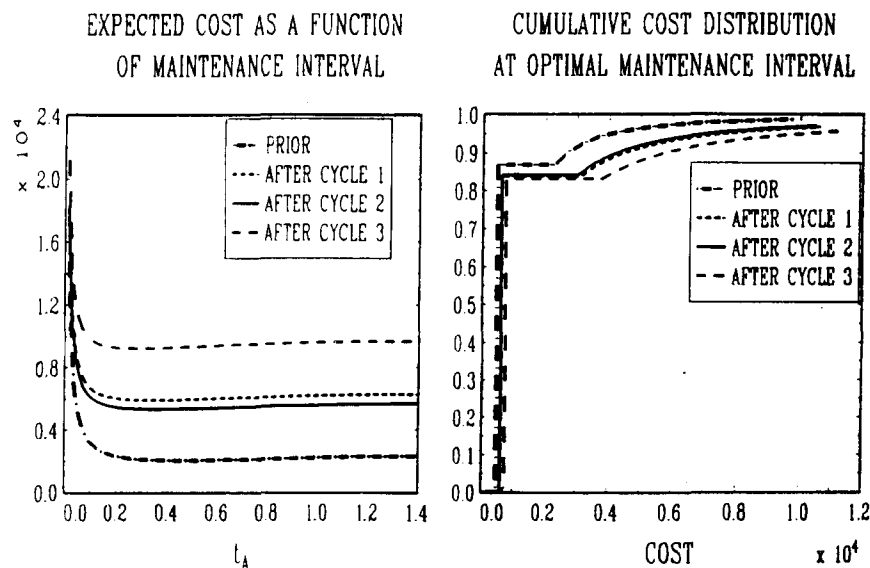


Fig. 4. Prior and posterior expected cost curves and cumulative cost distributions for age replacement.



posterior expected cost curves and cumulative cost distributions at the optimal maintenance interval. Again, the shift in expected cost curves and cost distributions imply a more pessimistic view of the air conditioners' performance as data is obtained. For this example, however, the expected cost at the optimal replacement interval provides an upper bound as it is approximately the 90th percentile of the cost distribution.

## CLOSING COMMENTS

The aim of this paper was to introduce the concept that optimal replacement and uncertainty analysis can be combined naturally by taking a Bayesian decision theoretic view of the replacement problem. The selection of both the Weibull process and Weibull distribution models as well as the priors was mainly for illustrative purposes. However, these models are central in the theory of repairable systems and reliability. Extensions of our approach to other failure models or the use of alternative prior distributions or utility functions are straightforward but may require use of numerical techniques. In such cases, computation of posterior distributions and other relevant quantities can be accomplished using Markov chain Monte Carlo methods such as Gibbs Sampling (see, for example, Gelfand & Smith).<sup>16</sup>

Extension of this approach to the original block replacement protocol and to the group replacement problem is currently under consideration.

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